

Challenge: Impute Missing Values with Mean

Mean imputation is a straightforward technique for handling missing values in numerical data. You replace each missing value in a column with the mean of the non-missing values from that same column. This method is most appropriate when the data is missing at random and the distribution of values is not heavily skewed. However, mean imputation can distort the variance and relationships in your data, especially if many values are missing or if the data is not normally distributed. It is important to consider these limitations before choosing mean imputation for your data cleaning workflow.

```
import pandas as pd
import numpy as np

data = {
    "id": [1, 2, 3, 4, 5],
    "score": [85, np.nan, 78, np.nan, 92]
}
df = pd.DataFrame(data)
print(df)
```